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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/152,569	01/10/2023	Jian-Hong LUO	AOIC027	4429

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APPLIED OPTOELECTRONICS, INC.
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SUGAR LAND, TX 77478

EXAMINER	
TAVLYKAEV, ROBERT FUATOVICH	

ART UNIT	PAPER NUMBER
2896	

NOTIFICATION DATE	DELIVERY MODE
09/11/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/152,569	Applicant(s) LUO et al.	
	Examiner ROBERT TAVLYKAEV	Art Unit 2896	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 5/20/25.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1 and 3-11 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1 and 3-11 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

DETAILED ACTION

3. Applicant's amendments and remarks filed 5/20/25 are acknowledged. Claim 1 has been amended, claim 2 canceled, and claim 11 added. Claims 1 and 3 – 11 are pending.

Response to Amendments / Arguments

4. Applicant's amendments have obviated the previously raised objection to claim 2.

5. Applicant's arguments regarding the amended claims versus the previously-raised rejections under 35 USC 103(a) have been fully considered but they are moot in view of the new grounds of rejections, as necessitated by Applicant's amendments. Specifically, the new limitations in independent claim 1 define that the accommodation through opening extends through the circuit board. As correctly noted by Applicant (p. 5 of the Remarks), Zhou does not teach that any of the two grooves in Figs. 1 – 4 extends through the circuit board 3. In fact, Figs. 2 and 4 show that the smaller of the two grooves (for mounting a PD array chip 13) does not extend through the circuit board 3. Accordingly, the Examiner has applied a reference by Kubo et al (US 2021/0278614 A1) that has been yielded by an updated prior art search and discloses

(Fig. 3) an optical module comprising a circuit board 130 with an opening an accommodation through opening 130a extending through the circuit board 130, and a transmitter optical subassembly including an optical modulator 154, wherein the modulator 154 is disposed in the accommodation through opening 130a. In combination with other prior art of record, Kubo teaches expressly or renders obvious all of the limitations recited by the amended claims, as detailed below.

Claim Rejections - 35 USC § 103

6. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

7. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the

contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

8. Claims 1, 3 – 6, 8, 9, and 11 are rejected under 35 U.S.C. 103 as being unpatentable over Kubo et al (US 2021/0278614 A1) in view of Zhou et al (CN 114706175 A), and further in view of "Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages" by Wang et al, Nature, vol. 562, pp. 103 – 112, 2018 (hereinafter Wang).

Regarding claim 1, Kubo discloses (Figs. 1, 2, 4, 6, and 7; Abstract; para. 0021 – 0042) an optical module (transceiver) configuration, comprising:

a circuit board 130 ("The board 130 is a circuit board" at para. 0024) comprising an accommodation through opening 130a extending though the circuit board 130 ("in the board 130, a through hole 130*a* is formed" at para. 0032); and

a transmitter optical subassembly (TOSA) 150 (para. 0021), comprising a light emitting unit 155 (laser; para. 32) and an opticla modulator 154, the optical modulator 154 being optically coupled with the light emitting unit 155 (to receive light from the laser and modulate it), and the optical modulator 154 being disposed in the accommodation through opening 130a (as shown in Figs 4 and 6) ("The TOSA 150 includes a metal plate 151 at its bottom portion, and on a lower surface thereof that is a component mounted surface of the metal plate 151, optical components 153 such as a lens and a ferrule, an optical modulator 154, light sources 155 such as laser diodes (LD), and a driver 156 are mounted. In response to a transmission signal supplied from the driver 156, the optical modulator 254 modulates light emitted by each light source 155" at para. 0033; "The optical components 153, the optical modulator 154, the light sources 155, and the

driver 156 mounted on the component mounted surface of the metal plate 151 are arranged inside the through hole 130*a* formed in the board 130" at para. 0034).

While Kubo does not detail a wide variety of optical modulators that are known in the art and commonly use in transceiver modules, Zhou discloses (Figs. 1 – 5; Abstract; para. 32 - 38) an optical module configuration (transceiver which is the same type as that in Kubo), comprising:

a circuit board 3,7 comprising an accommodation opening/groove (with a substrate 7 disposed over the opening, as seen in Fig 3; claim 6); and

a transmitter optical subassembly 4-6,8-10, comprising a light emitting unit 5 (laser; para. 32) and a *thin film lithium niobate (LiNbOx) modulator* 10 (para. 32), the thin film LiNbOx modulator 10 being optically coupled with the light emitting unit 5 (via 6 and 8, as shown in Figs. 3, and 5; para. 33), and *the thin film LiNbOx modulator* 10 being disposed in the accommodation opening/groove (a shown in Figs. 1 and 2; Abstract; claim 6).

It would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention that the optical modulator in Kubo can be a *thin film lithium niobate (LiNbOx) modulator* as a suitable/workable modulator type that is expressly taught by Zhou, provides efficient modulation at high speeds, and can enable 400G modules (Abstract; 1st para. of Section "Disclosure of Invention").

As seen in Fig. 6 of Kubo and Figs. 1 and 2 of Zhou, the accommodation through opening allows placement of optical modulators (of the thin film LiNbOx type) with different sizes up to a maximum size/footprint determined by a transverse size(s) of the accommodation through opening. While the Kubo – Zhou combination does not cite/exemplify a wide variety of thin film LiNbOx modulators supportive of different optical module specifications, such modulator variety is well known in the art. For example, Wang describes (Fig. 1; Abstract; pp.

101 – 102) a thin film LiNbOx modulator and provides examples (Table 1) of thin film LiNbOx modulators supportive of different optical module specifications (e.g., different 3dB electro-optic bandwidth of 45 GHz, 80 GHz, and 100 GHz in the first 3 rows of Table 1). It would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention that the accommodation through opening in Kubo can allows placement of various thin film LiNbOx modulators supportive of different optical module specifications, which are illustrated by Wang, in order to allow upgraded versions of thin film LiNbOx modulators with increasing 3dB bandwidths and modulation speeds. Such trend is noted by both Zhou (para. 2) and Wang (p. 103).

Table 1 of Wang shows that modulators operating at different speeds have different efficiencies, as defined by the voltage-length product. Hence, the Kubo – Zhou – Wang combination renders obvious that different modulator can have different dimensions, i.e., different lengths and/or different widths, and can each be accommodated in the accommodation through opening as long as it has a larger size than that of a modulator to be disposed therein. Thus, the Kubo – Zhou – Wang combination renders obvious that a second thin film LiNbOx modulator can have a shorter length in a direction orthogonal to a central axis of the circuit board than a first thin film LiNbOx modulator, and that the accommodation through opening allows placement of one of the first and second thin film LiNbOx modulators.

In light of the foregoing analysis, the Kubo – Zhou – Wang combination teaches expressly or renders obvious all of the recited limitations.

Regarding claim 3, the Kubo – Zhou – Wang combination considers (Fig. 4 of Kubo) that the circuit board 130,151 comprises a main body 130 comprising the accommodation through opening 130a and a base (lower portion of 151) disposed in the accommodation through

opening 130a (as seen in Fig. 4), the light emitting unit 155 and the optical modulator 154 (of the thin film LiNbOx type, according to Zhou) is carried on the base (as seen in Fig. 4). The base 7 has a smaller thickness than a total thickness and a total length of the accommodation through opening 130a (as seen in Fig. 4; "The optical components 153, the optical modulator 154, the light sources 155, and the driver 156 mounted on the component mounted surface of the metal plate 151 are arranged inside the through hole 130*a* formed in the board 130. The height of the lower surface of the driver 156 and the height of the lower surface of the board 130 are set substantially the same by adjusting the thickness of the metal plate 151, and the driver 156 is connected to the board 130 by wire bonding with a wire 157" at para. 0034).

Regarding claim 4, the Kubo – Zhou – Wang combination considers that the base (lower portion of the plate 151) can be made of a material with high thermal conductivity, such as a metal material ("As the metal plate 151, a copper alloy such as copper tungsten or copper molybdenum having high thermal conductivity, for example, can be used" at para. 0033 of Kobu).

Furthermore, the Examiner took official notice in the Office Action of 2/21/25 that it was well known in the art of optical transceiver modules that their heat sinks are commonly made of metal in order to ensure high thermal conductivity and efficient heat dissipation. Since Applicant has not traversed the official notice, the fact of common knowledge has become applicant admitted prior art. The use of the base as a heat sink would be obvious to a person of ordinary skill in the art.

Regarding claim 5, the Kubo – Zhou – Wang combination considers (Fig. 6 of Kubo) that the accommodation through opening 130a is non-coaxial with a central (longitudinal) axis of the circuit board 130.

Regarding claim 6, the Kubo – Zhou – Wang combination considers that the thin film LiNbOx modulator 10 can have a relatively short length, e.g., down to 5 mm (Extended Data Fig. 1c; 2nd para. of Section "Methods / Device fabrication" of Wang). Thus, the Zhou – Wang

combination considers renders obvious that a length of the accommodation through opening can be on the same order, that is from several millimeters to a few centimeters in order to accommodate both the thin film LiNbOx modulator 10 and the laser source 5 (as shown in Figs. 9 and 10 of Zhou). Hence, the Kubo – Zhou – Wang combination considers a range of transverse sizes of the accommodation through opening that at least overlaps with the recited range and a prima facie case of obviousness exists (MPEP 2144.05).

It is also noted that (i) the transverse sizes of the opening depend on a particular application (a maximum speed of operation, etc); that (ii) the instant application does not provide any criticality for the exact values of the recited suzes; that (iii) it has been held that discovering the optimum or workable ranges of prior art involves only routine skill in the art (In re Aller, 105 USPQ 233); and that (iv) it has been held that "A recognition in the prior art that a property is affected by the variable is sufficient to find the variable result-effective." *In re Applied Materials'*, Inc., 692 F.3d 1289, 1297 (Fed. Cir. 2012). It is well settled that it would have been obvious for an artisan with ordinary skill to develop workable or even optimum ranges for result-effective parameters. *In re Boesch*, 617 F.2d 272, 276 (CCPA 1980); see also *In re Woodruff*, 919 F.2d 1575, 1577-78 (Fed. Cir. 1990). In this regard, the Zhou – Wang combination considers the transverse sizes of the opening a result-effective parameter than enables placement of the thin film LiNbOx modulator 10 and the laser source 5 of certain sizes.

Regarding claim 8, the Kubo – Zhou – Wang combination considers (Figs. 1, 2, and 5 – 10 of Zhou) that the transmitter optical subassembly further comprises an output waveguide 11 (an optical fiber array 11; para. 32) disposed in the accommodation through opening (as seen in Figs. 1 and 2), and that the output waveguide 11 is optically coupled with a light propagation end of the thin film LiNbOx modulator 10 (as seen in Figs. 1, 2, and 5 – 10; para. 32 and 33).

Regarding claim 9, the Kubo – Zhou – Wang combination considers (Figs. 1, 2, and 5 – 10 of Zhou) that the thin film LiNbOx modulator 10 comprises a light receiving end (receiving light form the laser 5) and a light propagation end (emitting light to an optical fiber array 11; para. 32) at the same side (left side of 10 in Figs. 2 and 5 – 10).

Regarding claim 11, the Kubo – Zhou – Wang combination considers that the accommodation through opening can be rectangular shaped (as in Fig. 6 of Kubo; Fig. 2 of Zhou), while a hear sink and plate can have a different shape, e.g., an L-shape in Fig. 3 of Zhou. Thus, the Kubo – Zhou – Wang combination render obvious that the accommodation through opening can be shaped to comprises a first accommodation space and a second accommodation space arranged along a thickness direction of the circuit board, the first accommodation space (of a rectangular shape) and the second accommodation space (of an L-shape) have different horizontal widths, the thin film LiNbOx modulator is in the first accommodation space, and the base is in the second accommodation space.

9. Claims 7 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over Kubo in view of Zhou, in view of Wang, and further in view of Reimer et al (US 12,032,211 B1).

Regarding claim 7, the Kubo – Zhou – Wang combination considers that the light source (laser 155 in Kubo; laser 5 in Zhou) is optically coupled to a light receiving end of the thin film LiNbOx modulator 10 via *free-space* optics 6 (Fig. 5 of Zhou) or direct butt-coupling (Figs. 9 and 10 of Zhou) and does not teach that optical coupling can alternatively be provided by an optical *waveguide*. However, Reimer discloses (Figs. 1 and 2; Abstract; 4:7 – 7:47) an input (low-confinement) waveguide 120 (5:11 – 19) that is optically coupled with a light receiving end of a thin film LiNbOx modulator 112,114 (4:50 – 52; "The low-confinement waveguide is

optically coupled with the high-confinement waveguide" in the Abstract). It would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention that the light receiving end of the thin film LiNbOx modulator 10 can be optically coupled to the laser via an input waveguide, in accordance to the teachings of Reimer. The motivation for using an input waveguide is that it can be configured as a *mode-size transformer* and thereby reduce coupling loss between two optical elements with different sizes of optical modes, such as a laser (or an optical fiber) and a thin film LiNbOx modulator (1:25 – 38).

The Kubo – Zhou – Wang – Reimer combination considers that at least a portion (proximal to the thin film LiNbOx modulator 10) of the input waveguide is disposed in the accommodation through opening.

Regarding claim 10, the Kubo – Zhou – Wang – Reimer combination considers that light into the thin film LiNbOx modulator 10 can be provided either from a laser array (para. 33 of Zhou) or an optical fiber array (as shown for an optical fiber 150 in Fig. 1 of Reimer). In the latter case, the transmitter optical subassembly further comprises a (input) fiber array disposed in the accommodation through opening, and the (input) fiber array is optically coupled with a light receiving end of the thin film LiNbOx modulator 10 (in similarity with the output fiber array 11 in Figs. 1, 2 and 5 of Zhou).

Conclusion

10. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

11. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ROBERT TAVLYKAEV whose telephone number is (571)270-5634. The examiner can normally be reached 10:00 am - 6:00 pm, Monday - Friday.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, William Kraig can be reached on (571)272-8660. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ROBERT TAVLYKAEV/
Primary Examiner, Art Unit 2896